SUPERSESSION
(see Section 6)

MILITARY SPECIFICATION

FUZE, BOMB, ELECTRONIC, FMU-139/B

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the FMU-139/B Electronic Bomb Fuze, referred to herein as the "fuze."

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

UU-T-81 Tag, Shipping and Stock.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1325

AMSC NO. N3929

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MILITARY

MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1167	Ammunition Data Card.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOCUMENTS

Code of Federal Regulations (CFR)

49 CFR 100-199	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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DRAWINGS

United States Air Force

817274	Shipping and Storage, FMU-139/B Fuze Assembly (USAF), Packing and Marking.
8385215	FMU-139/B Fuze Assembly (USAF).

Naval Air Systems Command
(Code Ident 30003)

1379AS100	Fuze, FMU-139/B.
1379AS300	Shipping and Storage, FMU-139/B Fuze Assembly (USN), Packing and Marking.
1379AS500	Fuze, FMU-139/B Partially Loaded Test Sample.
1379AS701	FMU-139/B Fuze Schematic Diagram.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

(Copies of specifications, standards, drawings, publications and other Governmment documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The fuze is a joint service (Navy and Air Force), common bomb fuze used in the MARK 80 series and the M117 General Purpose Bombs (see 6.1 and 6.5). The fuze contains an explosive train mechanically and electrically out-of-line until specific weapon launch cycle events have occurred, after which it arms the warhead. The fuze provides arming times which may be selected either by the Fuze Function Control Set (FFCS) (for the majority of Navy applications), or by ground selection (for Air Force applications). The fuze can be initiated by either an external proximity sensor or the internal impact sensor.

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3.2 Performance characteristics.

3.2.1 Electrical. The electrical schematic is shown on Drawing 1379AS701.

3.2.1.1 Fuze resistance. The fuze shall have the following resistances measured at the connector:

a. 100 ohms ±20 percent between pins 3 and 2.

b. 120 kilohms ±20 percent between pins 3 and 1.

c. 20 megohms or greater between pins 1, 2, 3, and 4, connected together and the fuze housing.

d. Same as c when the measuring source polarity is reversed.

3.2.1.2 Navy mode. The fuze shall abort when power is applied in the Navy mode for a period of time that is greater than 500 milliseconds (ms).

3.2.1.2.1 Arming times with FFCSs. The fuze shall provide automatic selection of arming times of 2.6 seconds for the retarded mode, 5.5 seconds with a negative FFCS signal and 10.0 seconds with a positive FFCS signal in the unretarded mode. Accuracy of arming times shall be within ±5 percent over the temperature range from -40 to +160 degrees Fahrenheit (°F) (see Table I).

3.2.1.3 Navy mode interrupted power. The fuze shall abort when power is applied in the Navy mode for a time less than 500 ms and reapplied prior to expiration of arming time.

3.2.1.4 Air Force mode. The fuze shall abort when power is applied in the Air Force mode and the low drag arm time switch is in the "X" position.

3.2.1.4.1 Air Force arming times. Ground selectable options shall be 4.0, 6.0, 7.0, 10.0, 14.0, or 20.0 seconds. Ground selectable options shall be 2.0, 2.6, 4.0, or 5.0 seconds high drag arm time. Ground selection of the 2.0 seconds high drag arm shall require a positive action. Accuracy of arming times shall be ±5 percent over the operating temperature range of the fuze (see Figure 1 and Table I).

3.2.1.5 Delay times. The fuze shall provide three functioning delay times of 10 ±2, 25 ±5, or 60 ±12 ms plus instantaneous functioning with or without FFCSs (see Figure 1 and Table I).

3.2.2 Mechanical.

3.2.2.1 Retard sensor, low drag. When subjected to accelerations of less than 1.9 gravity units (g) applied at the midpoint of the sensor, the fuze shall select a low drag arming time.

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TABLE I. Arming and firing requirements.

Test	Input		Fuze setting 2/		Trigger 3/ source	Arming/firing requirements	Classification of defect
	Navy mode (Vdc) ±5%	Air Force mode (turbine) 1/	Low drag arm time (sec)	High drag arm/delay (sec/ms)			
1a	4/ 5/	High speed	X	2.6/Inst	Prox	Shall not arm or fire	CRITICAL
1b	-195 5/	...	X	2.6/Inst	Prox	Shall not arm or fire	CRITICAL
2	6/	Low speed	14	5.0/Inst	Prox	Shall not arm or fire	CRITICAL
3	...	High speed	20	5.0/Inst	Prox	Shall arm and fire	MAJOR
4	+300	...	X	2.6/25	Impact	Shall arm and fire	MAJOR
5	...	Low speed	6	4.0/Inst	Impact	Shall arm and fire	MAJOR
6	-195	...	X	2.6/60	Impact	Shall arm and fire	MAJOR
7	...	High speed	7	4.0/25	Prox	Shall arm and fire	MAJOR
8	-300	...	X	2.6/Inst	Prox	Shall arm and fire	MAJOR
9	...	Low speed	4	2.6/60	Impact	Shall arm and fire	MAJOR

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TABLE I. Arming and firing requirements. - Continued.

Test	Input		Fuze setting 2/		Trigger 3/ source	Arming/firing requirements	Classification of defect
	Navy mode (Vdc) ±5%	Air Force mode (turbine) 1/	Low drag arm time (sec)	High drag arm/delay (sec/ms)			
10	+195	...	X	2.6/10	Impact	Shall arm and fire	MAJOR
11	...	High speed	20	2.6/Inst	Impact	Shall arm and fire	MAJOR
12	-195	...	X	2.6/60	Prox	Shall arm and fire	MAJOR
13	...	Low speed	4	2.6/10	Prox	Shall arm and fire	MAJOR
14	+300	...	X	2.6/25	Prox	Shall arm and fire	MAJOR
15	...	High speed	10	2.6/25	Impact	Shall arm and fire	MAJOR
16	-300	...	X	2.6/Inst	Impact	Shall arm and fire	MAJOR
17	...	Low speed	6	4.0/25	Prox	Shall arm and fire	MAJOR
18	...	High speed	7	2.0/Inst	Prox	Shall arm and fire	MAJOR
19	...	High speed	10	4.0/Inst	Prox	Shall arm and fire	MAJOR

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TABLE I. Arming and firing requirements. - Continued.

Test	Input		Fuze setting 2/		Trigger 3/ source	Arming/firing requirements	Classification of defect
	Navy mode (Vdc) ±5%	Air Force mode (turbine) 1/	Low drag arm time (sec)	High drag arm/delay (sec/ms)			
20	...	High speed	14	2.6/Inst	Prox	Shall arm and fire	MAJOR
21	...	High speed	4	2.0/Inst	Prox	Shall arm and fire	MAJOR
22	+195	...	X	2.6/10	Prox	Shall arm and fire	MAJOR

1/ See 6.3.5 and 6.3.6 for definitions.

2/ See Figure 1. Fuze shall arm within ±5 percent of selected arm time setting. Early arm is a CRITICAL failure. Late arm is a MAJOR failure.

3/ Trigger source is defined as follows:

Prox: 40 ±5 microsecond pulsewidth and 15 ±2 volt amplitude at the proximity input with pin 2(-) and pin 3(+).

Impact: Shock level sufficient to provide 100 g minimum to the impact switches.

4/ This is not a destructive test. Subsequent to this test, subject the fuze to test 1b in Table I.

5/ Arming wire shall be in place.

6/ Safing pin shall be in place.

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3.2.2.2 Retard sensor, high drag. When subjected to an acceleration of 2.3 g or greater applied at the midpoint of the sensor, the fuze shall select a high drag arming time.

3.2.2.3 Impact sensors. The fuze shall not arm if an impact of 80 g or greater is sensed between 1.8 seconds and the selected arm time. After arming, the fuze shall fire when subjected to an impact of 80 g or greater.

3.2.3 Environmental. The fuze shall not be damaged nor shall safety or subsequent performance be degraded by exposure to the environments specified in 3.2.3.1 through 3.2.3.11. For the following requirements and the requirements of 3.2.4 through 3.2.4.3 definitions of the terms "safe" and "operable" shall be in accordance with MIL-STD-331.

3.2.3.1 Temperature and humidity. The fuze shall be safe and operable after exposure to the environment of Test 105 in MIL-STD-331.

3.2.3.2 Transportation vibration-temperature. The fuze shall be safe and operable after exposure to the environment of Test 119 in MIL-STD-331.

3.2.3.3 Vibration. The fuze shall be safe and operable after exposure to the environment of Method 514.3 in MIL-STD-810.

3.2.3.4 Salt fog. The fuze shall be safe and operable after exposure to the environment of Test 107 in MIL-STD-331.

3.2.3.5 Thermal shock. The fuze shall be safe and operable after exposure to the environment of Test 113 in MIL-STD-331.

3.2.3.6 Waterproofness. The fuze shall be safe and operable after exposure to the environment of Test 108 in MIL-STD-331, and there shall be no evidence that water entered the fuze cavity during exposure.

3.2.3.7 High temperature. The fuze shall be safe and operable after exposure to the environment of Method 501.2 in MIL-STD-810.

3.2.3.8 Low temperature. The fuze shall be safe and operable after exposure to the environment of Method 502.2 in MIL-STD-810.

3.2.3.9 1.5-meter drop. The fuze shall be safe and operable after exposure to the environment of Test 111 in MIL-STD-331.

3.2.3.10 Extreme temperature storage. The fuze shall be safe and operable, and no components or materials shall in any way be deformed, changed, or otherwise altered in such a manner as to degrade performance or safety after exposure to the environment of Test 112 in MIL-STD-331.

3.2.3.11 Leak. The fuze shall not leak at a rate greater than 1.0×10^{-2} standard cubic centimeters per second.

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3.2.4 Safety.

3.2.4.1 Jolt. The fuze shall not arm and shall be safe to handle, store, and transport, but need not be operable, after exposure to the environment of Test 101.1 of MIL-STD-331. No explosive component shall initiate during exposure. If the fuze is operable, the fuze shall not arm and shall remain safe when subjected to the tests of 4.6.1.

3.2.4.2 Twelve-meter drop. The fuze shall be safe to handle and dispose of after exposure to the environment of Test 103 of MIL-STD-331. No explosive component shall initiate during exposure.

3.2.4.3 Static detonator safety. There shall be no evidence of initiation, burning, charring, scorching, melting, or rupturing of the explosive lead when the detonator is initiated with the arming rotor in the safe position (see 6.3.3) after exposure to the environment of Test 115 of MIL-STD-331. Smudging of the surface of the explosive lead caused by deposits resulting from initiation of the detonator shall be acceptable. The fuze housing shall not rupture after exposure to the environment of this test.

3.2.5 Arming and firing. When arming and firing in accordance with the conditions specified in Table I, the fuze shall either, as applicable, not arm or initiate the explosive train, or shall arm within ±5 percent of the selected arm time setting and shall initiate the explosive train as follows:

a. Early initiation of any explosive is classified CRITICAL. Late or noninitiation of any explosive is classified MAJOR.

b. For partially loaded fuzes (see 4.5d) initiation of the explosive train shall be evidenced by detonation of the electric detonator.

c. For fully loaded fuzes (end-item fuzes in accordance with 3.4.1), initiation of the explosive-train shall be evidenced by detonation of the booster pellet.

3.2.5.1 Operating life. The fuze operating life shall be not less than 60 seconds. In the Air Force mode, the fuze shall meet this requirement at any temperature from -65 to +160°F. In the Navy mode, the fuze shall meet this requirement at any temperature from -40 to +160°F.

3.2.6 Visual. The fuze shall meet the following requirements when visually inspected:

a. There shall be no evidence of external damage, such as cracks, dents, or other blemishes, which may interfere with functioning or fit of the fuze at the next assembly.

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b. As shown on Figure 2, the words "EEDs ONLY" shall appear within a yellow band around the fuze housing of each partially loaded test sample fuze to indicate that the fuze contains electro-explosive devices (EEDs) only. In addition, a test sample tag in accordance with Figure 2 shall be attached to the fuze.

c. The fuze shall be in the safe position (see 6.3.1).

d. The low drag arm time switch shall be set to the "X" position and the high drag arm/delay switch shall be set to the "4.0/25" position.

e. The electrical connector shipping plug shall be properly installed on the fuze (see Drawing 1397AS100).

f. The warning tag assembly shall be properly installed.

3.2.7 Flight. The fuze shall meet the performance requirements specified in the flight test plan during exposure to actual flight environments of the flight test plan.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.2.3).

3.3.1 First article sample. The first article sample shall be composed of the following:

a. A sample of 213 fully loaded end-item fuzes in accordance with 3.4.1.

b. A sample of 63 partially loaded test sample fuzes, which shall be composed of the following:

(1) 54 fuzes with housing crimped.

(2) Nine fuzes without fuze housing crimped and with modification to permit firing.

3.4 Design and construction.

3.4.1 Production drawings. The fuze shall meet all requirements of and shall be manufactured and assembled in accordance with Drawing 1379AS100 and all associated drawings and documents.

3.4.2 Standards of manufacture.

3.4.2.1 Working environment and product cleanliness. The working environment shall meet the minimum levels and requirements of MIL-STD-1695.

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3.4.2.2 Soldering. Procedures and requirements for preparation and soldering of electrical connections shall be in accordance with WS-6536.

3.4.2.3 Workmanship. Fuze workmanship shall meet the requirements of MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein, and the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

(1) Individual unit acceptance inspections (see 4.3.1).

(2) Sampling inspection lot inspections (see 4.3.2).

4.2 First article inspections. When a first article inspection is required by the contract or purchase order, the first article inspections shall be performed by the contractor and the Government after award of contract and prior to production (see 6.2.3). First article inspections shall be performed on fuzes which have been produced with the equipment, materials, and procedures to be used in production. First article approval

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is valid only on the contract under which it is granted unless extended by the Government to other contracts. The Government reserves the right to require additional first article inspections until first article is approved and whenever regular production is interrupted for 120 days or more. Any production prior to approval of the first article shall be at the risk of the contractor (see 6.2.2).

4.2.1 First article inspection sequence. Prior to first article inspections, all fuzes shall have successfully completed the individual unit acceptance inspections of Figure 3 (see 4.3.1). The first article shall be subjected to the inspections of Figure 4 in the sequence shown. Inspections of Figure 4 shall be performed by the Government or by the contractor as specified in the contract or purchase order (see 6.2.3):

a. Laboratory tests: Nine fully loaded fuzes and 63 partially-loaded test sample fuzes (see 3.3) shall be shipped to the applicable test facility specified in the contract or purchase order for laboratory tests in the sequence shown on Figure 4.

b. Flight tests: 204 fully loaded fuzes (see 3.3) shall be shipped to the Government test facility specified in the contract or purchase order for flight tests (see 4.6.7).

4.2.2 First article acceptance and rejection criteria. Each first article fuze shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Figure 4, acceptance or rejection of the first article shall be based on the following criteria:

a. For flight tests, accept/reject criteria shall be in accordance with Figure 5.

b. For laboratory tests, defects in the fuze are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified herein; the first article shall be rejected if:

(1) One or more CRITICAL defects are found, or

(2) Six or more MAJOR defects of unidentical cause are found, or

(3) Two or more MAJOR defects of identical cause are found.

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NOTE: Although five MAJOR defects do not cause rejection of the first article, appropriate corrective action shall be taken for each MAJOR defect found to prevent recurrence prior to further inspection or production. Each defect shall be documented as specified in 4.5f.

c. The first article shall be rejected if, during any flight or laboratory test, the fuze arms early or if any explosive component of the fuze initiates early.

4.3 Quality conformance inspections.

4.3.1 Individual unit acceptance inspections. Individual unit acceptance inspections shall be performed in accordance with Figure 3 during manufacture of the fuze and prior to formation of the first article or the inspection lot. Fuzes found to be defective during the inspection sequence shall be rejected on an individual basis.

4.3.2 Sampling inspection lot. A sampling inspection lot shall consist of partially-loaded fuzes in accordance with 4.5d, fabricated under the same conditions and offered for acceptance inspection at the same time. For safety purposes, the inspection lot shall be formed with fuzes lacking leads and boosters (see 4.3.2.2).

4.3.2.1 Sampling inspection lot size. The actual sampling inspection lot size shall consist of not less than 281 nor more than 3,200 fuzes.

4.3.2.2 Sampling inspection sequence. Prior to formation of the inspection lot, all fuzes shall have passed the individual unit acceptance inspections of Figure 3 (see 4.3.1). After completion of inspections of Figure 3 and prior to installation of the leads and boosters, the inspection lot shall be formed. The required quantity of fuzes (see 4.3.2.2.1) shall then be randomly selected from the lot. The sampling inspection sample fuzes shall be subjected to the applicable inspections of Figure 6 in the sequence shown; the remainder of the lot shall be held in bond pending successful completion of the sampling inspection sequence. If the sample fuzes do not cause rejection, the remainder of the fuzes shall then be removed from bond, fully loaded (live leads and boosters installed), and the fuze housings shall be formed (crimped) in accordance with Drawing 1379AS100. The loading process shall be verified by performing the Process Control Test (PCT) specified in Appendix B.

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4.3.2.2.1 Sampling inspection lot. Sampling shall be as specified below.

a. For lot sizes of 281 to 1,200 fuzes, sampling shall be in accordance with MIL-STD-105, Single Sampling Plan, Special Inspection Level S-4, Normal Inspection, Sample Size Code Letter F, acceptable quality level (AQL) of 2.5 percent defective.

b. For lot sizes of 1,201 to 3,200 fuzes, sampling shall be in accordance with MIL-STD-105, Single Sampling Plan, Special Inspection Level S-4, Normal Inspection, Sampling Size Code Letter G, AQL of 2.5 percent defective.

4.3.2.3 Sampling inspection lot acceptance and rejection criteria. Each fuze shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Figure 6, acceptance or rejection of the fuzes shall be based on the following criteria:

a. Where 100-percent inspection is required, acceptance or rejection shall be on an individual basis.

b. For inspections on sample fuzes, defects in the fuze are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified herein. The fuzes shall be rejected if one or more CRITICAL defects are found. For MAJOR defects, acceptance and rejection shall be in accordance with the AQL specified in 4.3.2.2.1.

c. The sampling inspection lot shall be rejected if, during any test, the fuze arms early or if any explosive component of the fuze initiates early.

4.4 Measuring and test equipment.

4.4.1 Calibration and maintenance. All measuring and test equipment used by the contractor for performance of the inspections specified herein shall be calibrated and maintained in accordance with MIL-STD-45662. (see 6.2.2)

4.4.2 Standard test equipment. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements, under the conditions, and to the accuracies specified herein. Unless otherwise

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specified herein, the magnitude of any error introduced by the test equipment shall not exceed ten percent of the allowable tolerance of the requirement being measured. Any error introduced by the test equipment shall not be applicable toward widening the allowable tolerance of the requirement being measured.

4.4.3 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein or supplied by the Government, are devised or commercially available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.4.2 and are subject to verification by the contracting activity.

4.4.3.1 Electrical test set. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall design and fabricate an electrical test set for performance of the electrical inspections of 4.6.1. The test set shall meet the requirements of 4.4.2 and 4.4.3 and will be certified by the Government.

4.4.4 Gauges. The contractor shall provide whatever gauges are necessary to ensure conformance to the dimensional requirements of this specification and the applicable drawings. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contracting activity will furnish drawings of pertinent Government final inspection gauges. This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gauges required for fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gauges.

4.5 Test conditions and provisions. Unless otherwise specified herein, the following test conditions and provisions shall apply for performance of the inspection methods of 4.6:

a. Safety precautions. The fully-loaded fuze contains explosives as shown in Table II. The inspections of 4.6 require the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews (see 6.4).

b. Test environment. Unless otherwise specified herein, the inspection methods of 4.6 shall be performed under the conditions specified in 3.4.2.1.

c. Test fixtures. Unless otherwise specified in the contract or purchase order (see 6.2.1), all test fixtures for performance of the inspections of 4.6 shall be supplied by the contractor and shall be verified by the contracting activity prior to use.

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TABLE II. Fuze explosive contents.

Component	Type of explosive	Amount (weight)
MARK 71 MOD 1 Electric Detonator	Lead azide/PETN	80 milligrams lead azide 40 milligrams PETN
MARK 8 MOD 0 Explosive Lead	Composition CH-6	188 milligrams
MARK 20 MOD 0 Bellows Motor	LMNR and black powder	50 milligrams
	Lead styphnate	2-6 milligrams
Piston actuator	Lead styphnate	12 milligrams
Booster pellet	Composition CH-6	125 grams

d. Partially loaded test sample fuzes. The contractor shall prepare partially loaded test sample fuzes in the quantity specified herein for first article and sampling inspections. Partially loaded test sample fuzes shall be fuzes in accordance with Drawing 1379AS500, a test sample tag in accordance with Figure 2, shall be attached to all partially loaded fuzes and shall remain attached through final disposition.

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e. Inspection sequences. The inspections of 4.6 shall be performed in the sequences shown on Figures 3, 4 and 6, as applicable. When feasible, sequential inspections may be combined, provided it can be demonstrated by the contractor that the applicable requirements of Section 3 have been individually met.

4.6 Inspection methods. To determine conformance to the requirements of Section 3, the inspections specified in 4.6.1 through 4.6.1.5 shall be performed in accordance with the test conditions and provisions of 4.5.

WARNING

The fuze contains explosive components that are sensitive and hazardous. Suitable safety precautions shall be observed during handling and testing of the fuze (see 4.5a and 6.4).

4.6.1 Electrical (electronic diagnostic tests (EDTs)).

4.6.1.1 Resistance tests. To determine conformance to 3.2.1.1, test the fuze as follows:

a. Measure the resistance between input connector pins 3 and 2. The measurement source voltage shall be 0.00 to 0.20 volts direct current (Vdc). The resistance shall be 100 ohms ±20 percent.

b. Measure the resistance between input connector pins 3 and 1. The measurement source voltage shall be 0.00 to 0.20 Vdc. The resistance shall be 120 kilohms ± 20 percent.

c. Measure the resistance between input connector pins 1, 2, 3, and 4 connected together and the fuze metal structure or can. The measurement source voltage shall be 1.5 to 5.0 Vdc. The resistance shall be not less than 20 megohms.

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d. Reverse the polarity of the measurement source and repeat the measurement of step c. The measurement source voltage shall be 1.5 to 5.0 Vdc. The resistance shall be not less than 20 megohms.

4.6.1.2 Navy mode. To determine conformance to 3.2.1.2, test the fuze as follows:

a. Set the low drag arm time switch to the "X" position.

b. Set the high drag arm/delay switch to the "4.0/25" position.

c. Verify that the safing pin is properly installed (see Drawing 1379AS100).

WARNING

Do not proceed with the test if the safing pin is not properly installed.

d. Apply a voltage of 100 ±5 Vdc across pins 1(+) and 3(-) until the fuze aborts or for 1.0 second minimum.

e. Record the current waveform of the 100 Vdc power input. Examine the waveform to determine conformance to Figure 7.

Attempted arm: CRITICAL

4.6.1.3 Navy mode interrupted power. To determine conformance to 3.2.1.3, test the fuze as follows:

a. Set the low drag arm time switch to the "X" position.

b. Set the high drag arm/delay switch to the "4.0/25" position.

c. Verify that the safing pin is properly installed (see Drawing 1379AS100).

WARNING

Do not proceed with the test if the safing pin is not properly installed.

d. Apply a voltage of 100 ±5 Vdc across pins 3(+) and 1(-) for 30 ±10 ms and then reapply power at 1.2 ±0.1 seconds later until the fuze aborts or for 5.9 seconds minimum.

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e. Record the current wave form of the 100 Vdc power input. Examine the wave form to determine conformance to Figure 8.

Attempted arm: CRITICAL

4.6.1.4 Air Force mode. To determine conformance to 3.2.1.4 test the fuze as follows:

a. Set the low drag arm time switch to the "X" position.

b. Set the high drag arm/delay switch to the "4.0/25" position.

c. Verify that the safing pin is properly installed (see Drawing 1379AS100).

WARNING

Do not proceed with the test if the safing pin is not properly installed.

d. Simultaneously apply voltages of 5.0 ±0.5 Vdc and 100 ±5 Vdc across pins 4(-) and 3(+), and across pins 1(+) and 3(-), respectively (common leads to pin 3), until the fuze aborts or for 1.0 second minimum.

e. Record the current waveform of the 100 Vdc power input. Examine the waveform to determine conformance to Figure 9.

Attempted arm: CRITICAL

NOTE: The specified input shall be used instead of Table I requirements.

4.6.2 Mechanical.

4.6.2.1 Retard sensors, low drag. To determine conformance to 3.2.2.1, test the fuze as follows:

a. Verify that the safing pin is properly installed (see Drawing 1379AS100).

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WARNING

Do not proceed with the test if the safing pin is not properly installed.

b. Set the low drag arm time switch to the "X" position and set the high drag arm/delay switch to the "4.0/25" position.

c. Mount the fuze (see 4.5e) so acceleration will be applied along the "Z(-)" axis as shown on Figure 10.

d. Accelerate the fuze to a g level between 1.6 and 1.9 g at the midpoint of the sensor. While maintaining this acceleration level, apply a voltage of 100 ±5 Vdc across pins 3(+) and 1(-) of the fuze for 30 ±10 ms (see 6.3.1). At 1.2 ±0.1 seconds reapply power until the fuze aborts or for 5.9 seconds minimum. Examine the current waveform to determine conformance to Figure 8 and verify the selected arm time.

CRITICAL

4.6.2.2 Retard sensor, high drag. To determine conformance to 3.2.2.2, repeat the test of 4.6.2.1, except accelerate the fuze to a g level of between 2.3 and 3.5 g at the midpoint of the sensor. Examine the current waveform to determine conformance to Figure 11 and to verify the selected arm time. Remove the fuze from the fixture. Set the switch to the "X" position and the high drag arm/delay switch to the "4.0/25" position.

MAJOR

4.6.2.3 Impact sensors. To determine conformance to 3.2.2.3, test the fuze as follows:

a. Set the fuze low drag arm time switch to the "X" position.

b. Set the fuze high drag arm/delay switch to the "4.0/25" position.

c. Verify that the safing pin is properly installed (see Drawing 1379AS100).

WARNING

Do not proceed with the test if the safing pin is not properly installed.

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d. Apply a voltage of 100 ±5 Vdc across pins 3(+) and 1(-) for 30 ±10 ms and then reapply 1.2 ±0.1 seconds later.

e. At time 1.3 ±0.2 seconds after reapplication of power in step d, provide an impact shock to the fuze. Shock level shall be sufficient to provide 100 g minimum to the impact switches.

f. Record the current waveform of the 100 Vdc power input. Examine the waveform to determine conformance to Figure 12.

MAJOR

4.6.3 Environmental. Unless otherwise specified in the contract or purchase order (see 6.2.1), any fuze subjected to the environmental tests specified in 4.6.3.1 through 4.6.3.11 and the tests of 4.6.4 and 4.6.5 shall be considered destroyed and shall not be returned to the lot or forwarded for service use, but shall be labeled appropriately and held for disposal. Throughout the environmental tests, the fuze shall be in the safe position (see 6.3.1) unless otherwise specified. After exposure to the following environments, the fuze shall be subjected to operability tests in the sequence shown on Figure 4 or 6, as applicable.

4.6.3.1 Temperature and humidity. To determine conformance to 3.2.3.1, test the fuze in accordance with MIL-STD-331, Test 105, Single Chamber Cycling Method, except the low temperature extreme shall be -65 °F (instead of -80°F).

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.3.2 Transportation vibration-temperature. To determine conformance to 3.2.3.2, test the fuze in accordance with MIL-STD-331, Test 119, Procedure 2.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.3.3 Vibration. To determine conformance to 3.2.3.3, test the fuze as follows:

a. First article. Test the fuze in accordance with MIL-STD-810, Method 514.3, Equipment Category 7b, Procedure I, Figure 514.3-33, except that W_1 = .04 g^2/Hz, W_2 = .15 g^2/Hz, f_1 = 100 Hz and f_2 = 1000 Hz. Vibration shall be applied for 30 minutes in each axis. (The fuze shall not be operated during the test.)

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b. Sampling inspection. Test the fuze in accordance with MIL-STD-810, Method 514.3, Equipment Category 7b, Procedure I, Figure 514.3-33, except that W_1 = .04 g^2/Hz, W_2 = .13 g^2/Hz, f_1 = 100 Hz, f_2 = 1000 Hz. Vibration shall be applied for 30 minutes in each axis. The fuze shall not be operated during the test.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.3.4 Salt fog. To determine conformance to 3.2.3.4, test the fuze in accordance with MIL-STD-331, Test 107, 48-hour test. Following exposure, two fuzes shall be subjected to the sequential inspections of Figure 4 when wet; three fuzes shall be dried and then subjected to the sequential inspections of Figure 4.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.3.5 Thermal shock. To determine conformance to 3.2.3.5, test the fuze as follows:

a. First article. Test the fuze in accordance with MIL-STD-331, Test 113.

b. Sampling inspection. Test the fuze in accordance with MIL-STD-331, Test 113, except temperature cycling shall be between -60 and +140°F (instead of -65 and +165°F).

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.3.6 Waterproofness. To determine conformance to 3.2.3.6, test the fuze in accordance with MIL-STD-331, Test 108. There shall be no evidence that water has entered the fuze.

Remain safe: CRITICAL
Remain operable: MAJOR
No evidence of water in cavity: MAJOR

4.6.3.7 High temperature. To determine conformance to 3.2.3.7, test the fuze as follows:

a. First article. Test the fuze in accordance with MIL-STD-810, Method 501.2, Procedure II, steps 1, 7, 8, 9, 16 and 17. In step 7 the temperature is to be 160°F, relative humidity held to 15 percent or lower, and exposure time shall be 2 hours. The internal chamber temperature shall be raised to 220°F for

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the final 10 ±1 minutes of exposure. For fuzes tested in the Navy mode, the chamber temperature shall remain at 160°F for the last 10 ±1 minutes and shall not be raised to 220°F. In step 16, remove the fuze from the temperature chamber and perform the tests of 4.6.5.1 and 4.6.5.1.1 within 5 minutes after removal from the chamber.

b. Sampling inspection. Test the fuze in accordance with MIL-STD-810, Method 501.2, Procedure II, steps 1, 7, 8, 9, 16 and 17. In step 7, the temperature is to be 145°F, and the relative humidity held to 15 percent or lower. The exposure time shall be 2 hours. In step 16, remove the fuze from the temperature chamber and perform the tests of 4.6.5.1 and 4.6.5.1.1 within 5 minutes after removal from the chamber.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.3.8 Low temperature. To determine conformance to 3.2.3.8, test the fuze as follows:

a. First article. Test the fuze in accordance with MIL-STD-810, Method 502.2, Procedure II. Steps 6 and 7 shall be omitted. The low temperature in step 1 shall be -65°F. The exposure time shall be 2 hours. For fuzes tested in the Navy mode, the low temperature shall be -40°F. In step 4, remove the fuze from the temperature chamber and perform the tests of 4.6.5.1 and 4.6.5.1.1 within 5 minutes after removal from the chamber.

b. Sampling inspection. Test the fuze in accordance with MIL-STD-810, Method 502.2, Procedure II. Steps 6 and 7 shall not be performed. The low temperature in step 1 shall be -35°F. The exposure time shall be 2 hours. In step 4 remove the fuze from the temperature chamber and perform the tests of 4.6.5.1 and 4.6.5.1.1 within 5 minutes after removal from the chamber.

Remain safe: CRITICAL
Remain operable: MAJOR

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4.6.3.9 1.5-meter drop. To determine conformance to 3.2.3.9, test the fuze with safing pin installed in accordance with MIL-STD-331, Test 111, Procedure 2, Criteria B.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.3.10 Extreme temperature storage. To determine conformance to 3.2.3.10, test the fuze in accordance with MIL-STD-331, Test 112, Procedure I.

Remain safe: CRITICAL
Remain operable: MAJOR
No deformation or alteration of components: MAJOR

4.6.3.11 Leak. To determine conformance to 3.2.3.11, apply 15 ±2 pounds per square inch gage (psig) to the exterior surfaces of the fuze. Measure leak rate into the fuze.

MAJOR

4.6.4 Safety.

4.6.4.1 Jolt. To determine conformance to 3.2.4.1, test the fuze with safing pin installed in accordance with MIL-STD-331, Test 101, Procedure I. For the test with the "Z" axis (see Figure 10) horizontal to the shock table, orient the fuze in the test fixture so that the "X(-)" axis is normal to the impact surface.

CRITICAL

4.6.4.2 Twelve-meter drop. To determine conformance to 3.2.4.2, install the fuze in either the aft or the nose fuze well, as specified below and on Figure 10, of a test vehicle simulating a loaded MARK 82 Bomb. Test the fuze in accordance with MIL-STD-331, Test 103, Procedure I, except the orientation at impact shall be as follows:

a. One fuze shall be installed in the tail fuze well and dropped with the longitudinal axis of the test vehicle ("V" axis shown on Figure 10) vertical, nose up.

b. One fuze shall be installed in the tail fuze well and dropped with the longitudinal axis of the test vehicle vertical, nose down.

c. One fuze shall be installed in the tail fuze well and dropped with the longitudinal axis of the test vehicle horizontal.

d. One fuze shall be installed in the nose fuze well and dropped with longitudinal axis of the test vehicle horizontal.

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e. One fuze shall be installed in the tail fuze well and dropped with the longitudinal axis of the test vehicle 45 degrees from vertical, nose up.

CRITICAL

NOTE: For the twelve-meter drop test of 4.6.4.2, the fuze shall be mounted in the aft fuze well or the nose fuze well of the test vehicle with the "X(-)" axis of the fuze coincident with the "U" axis of the test vehicle. The bomb should be assembled with tail fin when dropped in accordance with 4.6.4.2.

4.6.4.3 Static detonator safety. To determine conformance to 3.2.4.3, test the fuze in accordance with MIL-STD-331, Test 115. The modification includes installation of a lead, and the booster is not required.

CRITICAL

NOTE: Modification of the fuze for the static detonator safety test shall be in accordance with the accepted test plan (see 4.5e).

4.6.5 Arming and firing.

4.6.5.1 Partially loaded fuzes. To determine conformance to 3.2.5, arm the fuze (see 6.3.2), measure the arming time, and initiate the fuze under the conditions specified in Table I, as applicable. For both the first article and inspection lot, one each sample fuze shall be subjected to one of the tests (one fuze per test) of Table I in the numerical sequence specified in the Table. The sequence need not be completed (Test 1 through 21), contingent upon the quantity of fuzes to be tested; for example, if 16 fuzes are to be tested, Tests 1 through 16 only shall be performed. However, if the quantity of fuzes to be tested exceeds the number of tests, as in the case of the first article, the sequence shall be repeated, beginning with Test 1, until all sample fuzes are exhausted.

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NOTE: Test equipment and procedures for performance of the arming and firing tests of Table I shall be in accordance with the accepted test plan (see 4.5e). The test plan shall also include details for determining, conclusively, conformance to the requirements of 3.2.5a and b. The classifications of defects are specified in Table I.

4.6.5.1.1 Operating life. To determine conformance to 3.2.5.1, perform step 16 of the high temperature test (see 4.6.3.7) and step 4 of the low temperature test (see 4.6.3.8) in accordance with the tests of Table I. Wait not less than 60 seconds before applying a proximity signal to fire the fuze. For first article fuzes, (see Figure 4), and for sampling inspection fuzes, (see Figure 6).

4.6.5.2 Fully loaded fuzes. To determine conformance to 3.2.5, arm the fuze (see 6.3.2). The low drag arming time switch shall be in the "X" position. The high drag arm/delay switch shall be in the "4.0/25" position, and the input power shall be -195 Vdc in the FFCS mode. The fuze shall be supplied with a proximity signal for firing. The arming time shall be measured.

WARNING

Use extreme caution at all times when handling and testing fully loaded fuzes. Failed fuzes shall be handled in accordance with approved safety procedures.

4.6.6 Visual. To determine conformance to 3.2.6, visually inspect the fuze for the following:

a. Evidence of external damage (see 3.2.6a).

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b. The words "EEDs ONLY" appear within a yellow band around the fuze housing of partially-loaded test sample fuze and test sample tag attached (see 3.2.6b).

CRITICAL

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c. Fuze in safe position (see 3.2.6c and 6.3.1).

CRITICAL

d. Fuze low drag arm time switch set to the "X" position and fuze high drag arm/delay switch set to the "4.0/25" position (see 3.2.6d).

CRITICAL

e. Electrical connector shipping plug properly installed (see 3.2.6e).

MAJOR

f. Warning tag assembly properly installed.

CRITICAL

4.6.7 Flight. To determine conformance to 3.2.7 and when specified in the contract or purchase order, the Government will perform flight tests on a sample quantity of fuzes from the first article in accordance with the test conditions specified in the flight test plan (see 6.2.3).

4.6.8 Drawing conformance. To determine conformance to 3.4.1, perform drawing conformance inspections on each lot of parts, subassemblies, and assemblies of the fuze during manufacture, and on the completely assembled fuze.

4.6.9 Packaging. To determine conformance to Section 5, inspect the packaging, packing, and marking of the fuze prior to shipment.

MAJOR

5. PACKAGING

5.1 Preservation. Preservation shall be in accordance with FED-STD-102, Level A, and as specified on Drawing 1379AS300 for Navy use or Drawing 817274 for Air Force use. Preservation shall conform to 49 CFR 100-199.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level A, and as specified on Drawing 1379AS300 for Navy use or Drawing 817274 for Air Force use. Navy shipping containers have spaces for nine fuzes, and Air Force shipping containers have spaces for six fuzes. Fuzes, packaged in accordance with 5.1, shall be packed in M548 shipping containers in accordance with Drawing 1379AS300 or Drawing 817274. Packing and containers shall be suitable for explosives and shall conform to 49 CFR 100-199.

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5.2.1 Navy ammunition data card. When specified in the contract or purchase order (see 6.2.2), one Navy ammunition data card, completed in accordance with MIL-STD-1167, and listing the explosive components, shall be packed inside each shipping container.

5.2.2 Air force ammunition data card. When specified in the contract or purchase order (see 6.2.2), one ammunition data card, completed in accordance with MIL-STD-1167, and listing the explosive components, shall be packed with each pallet.

5.3 Marking. Unless otherwise specified in the contract or purchase order (see 6.2.1), in addition to any special marking required by the contract or purchase order, all interior and exterior marking shall be in accordance with MIL-STD-129 and Drawing 1379AS300 for Navy use or Drawing 817274 for Air Force use.

6. NOTES

6.1 Intended use. The fuze is intended for use by the Navy and the Air Force in the MARK 80 Series and M117 General Purpose Bombs (see 3.1 and 6.5).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspections and facilities, if other than as specified in 4.1.

c. Design and fabrication of electrical test set by contractor, if required (see 4.4.3.1).

d. Government final inspection gauge drawings, if required (see 4.4.4).

e. Approval of test fixtures by contracting activity (see 4.5c).

f. Disposal of sample fuzes subjected to environmental and safety inspections (see 4.6.3).

g. Special marking, if required (see 5.3).

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.2	Test Plan	DI-T-5204	---
4.4.1	Calibration System Description	DI-R-7064	---
5.2.1	Ammunition Data Card	DI-MISC-80043	---
5.2.2	Data Cards for Explosive Assemblies, Sub-Assemblies, and Parts	DI-L-3317	---

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DOD 5010.12-L , AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of 213 fully-loaded fuzes and 63 partially-loaded fuzes. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product

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which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. The following should also be identified.

a. Identification of the test facility to perform the laboratory tests of Figure 4.

b. Identification of the Government test facility to perform flight testing, if required, and applicable details of flight tests, including performance requirements and test conditions (see 3.2.7, 4.2.1b, 4.7.2a, and 4.6.7).

c. Identification of first article spares.

d. Identification of a facility to perform a Physical Configuration Audit, if required by the contracting agency, and details of the audit.

6.3 Definitions.

6.3.1 Safe position. The fuze is considered to be in the safe position when the gag rod is flush or below the end of the gag rod/arming wire housing and no evidence of the color red is visible on the gag rod.

6.3.2 Armed position. The fuze is considered to be in the armed position when the gag rod extends beyond the gag rod/arming wire housing. The red-striped portion of the gag rod should be visible.

6.3.3 Navy mode. For the inspections specified herein, the fuze is considered to be in the Navy mode when the input voltage to the fuze is +195, -195, +300, or -300 Vdc applied for a period of time greater than 15 and less than 500 ms at pins 1 and 3 and when no power is applied across pins 4 and 3 (turbine release line).

6.3.4 Air Force mode. For inspections specified herein, the fuze is considered to be in the Air Force mode when the inputs to the fuze are as specified in either 6.3.5 for low-speed turbine or in 6.3.6 for high-speed turbine and when 5.0 ±0.5 Vdc is applied simultaneously across pins 4(+) and 3(-) (turbine release line).

6.3.5 Low-speed turbine. For the inspections specified herein, low-speed turbine is defined as when the inputs to the fuze are as follows:

a. Waveform: Half-wave rectified, sinusoidal.

b. Peak voltage: 40 ±2 Vdc.

c. Frequency: 1.0 ±0.1 kilohertz (kHz).

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6.3.6 High-speed turbine. For the inspections specified herein, high-speed turbine is defined as when the inputs to the fuze are as follows:

a. Waveform: Half-wave rectified, sinusoidal.

b. Peak voltage: 120 ±6 Vdc.

c. Frequency: 6.0 ±0.6 kHz.

6.3.7 Abort. A programmed response by the fuze which dissipates all internal electrical energy.

6.4 Explosive safety precautions. The manufacture, assembly, and inspection of both fully loaded fuzes and partially loaded test sample fuzes require loading and handling of explosive components which can be hazardous to personnel and equipment. These operations require the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews. Minimum explosives safety precautions for use by the contractor are detailed in DOD Instruction 4145.26M, "DOD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material."

6.5 FMU-139/B fuze assemblies. FMU-139/B fuze assemblies are comprised as follows:

a. For Navy use. The FMU-139/B Fuze Assembly (USN) of Drawing 1379AS400 will consist of one FMU-139/B Electronic Bomb Fuze in accordance with Drawing 1379AS100, and one closure ring in accordance with Drawing 1379AS198.

b. For Air Force use. The FMU-139/B Fuze Assembly (USAF) of Drawing 8385215 will consist of one FMU-139/B Electronic Bomb Fuze in accordance with Drawing 1379AS100, one closure ring in accordance with Drawing 1379AS198, one FZU-48/B Bomb Fuze Initiator in accordance with Drawing 8385220, and one power cable assembly in accordance with Drawing 8385212.

6.6 Subject term (key word) listing.

Bomb Fuzes
Electronic Fuzes
Explosive Trains
Explosives
FMU 139/B Electronic Bomb Fuze
Free Fall Weapons
Fuzes (Ordnance)
Impact Fuzes

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6.7 Changes from previous issue. Symbols are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the change.

6.8 Supersession data. This specification supersedes the following:

(a) MIL-F-85571(AS), dated 4 February 1985.

(b) ECP 30-103 (NOR 30-103.1R1) dated 26 August 1985.

(c) ECP 30-059 (NOR 30-059.1R1) data 28 August 1985.

Preparing activity:
Navy - AS
(Project 1325-N127)

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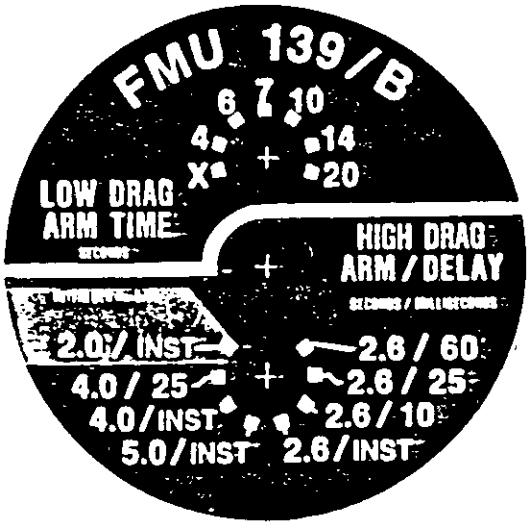
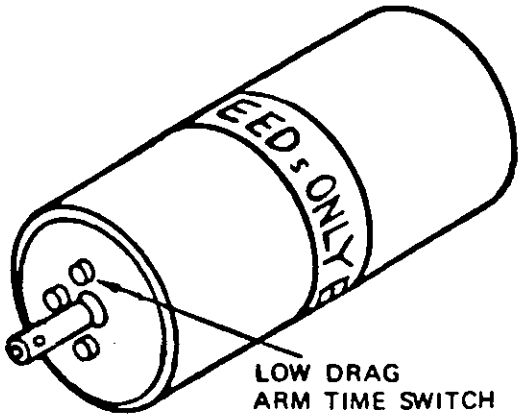
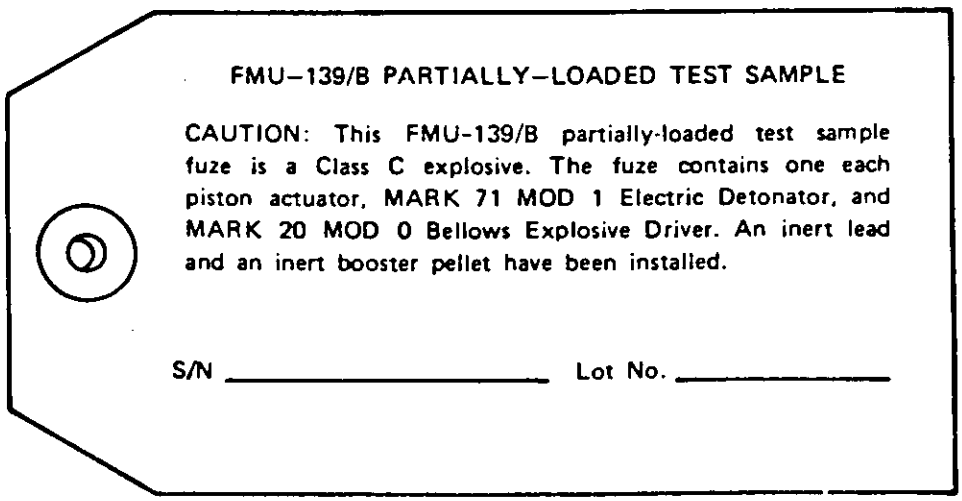


FIGURE 1. Front panel label.

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NOTE: The material for marking shall be plastic, pressure-sensitive adhesive tape in accordance with MIL-P-38477, Type I, Class 2, 1-inch roll stock, pigmented in accordance with FED-STD-595, Color No. 23538. The words "EEDs ONLY" shall be indelibly printed on the tape in .375 to .500 inch black letters, block style. The tape shall be positioned around the circumference of the fuze housing, with the words "EEDs ONLY" repeating a minimum of 3 times around the circumference.



NOTE: The tag shall be in accordance with UU-T-81, Type B, Grade 15CSU, Size 7 or 8, Color red. The information may be printed or stamped thereon.

FIGURE 2. Marking and tag for partially-loaded test sample fuze.

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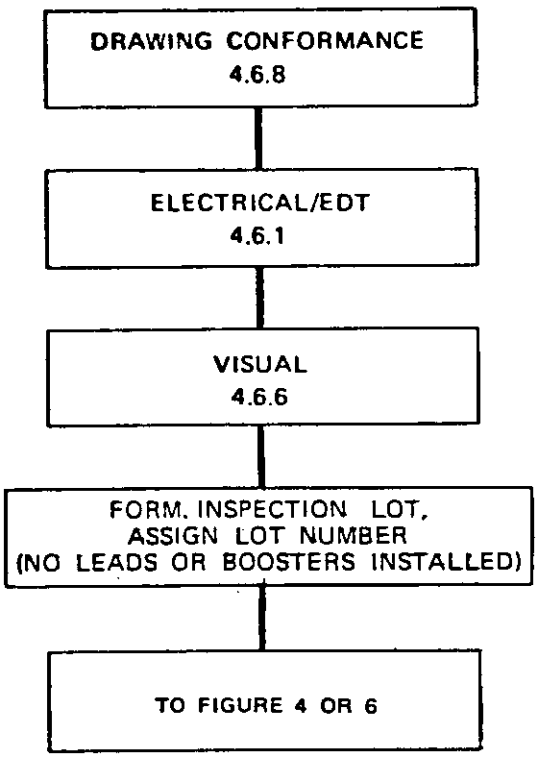


FIGURE 3. Individual unit acceptance inspection sequence.

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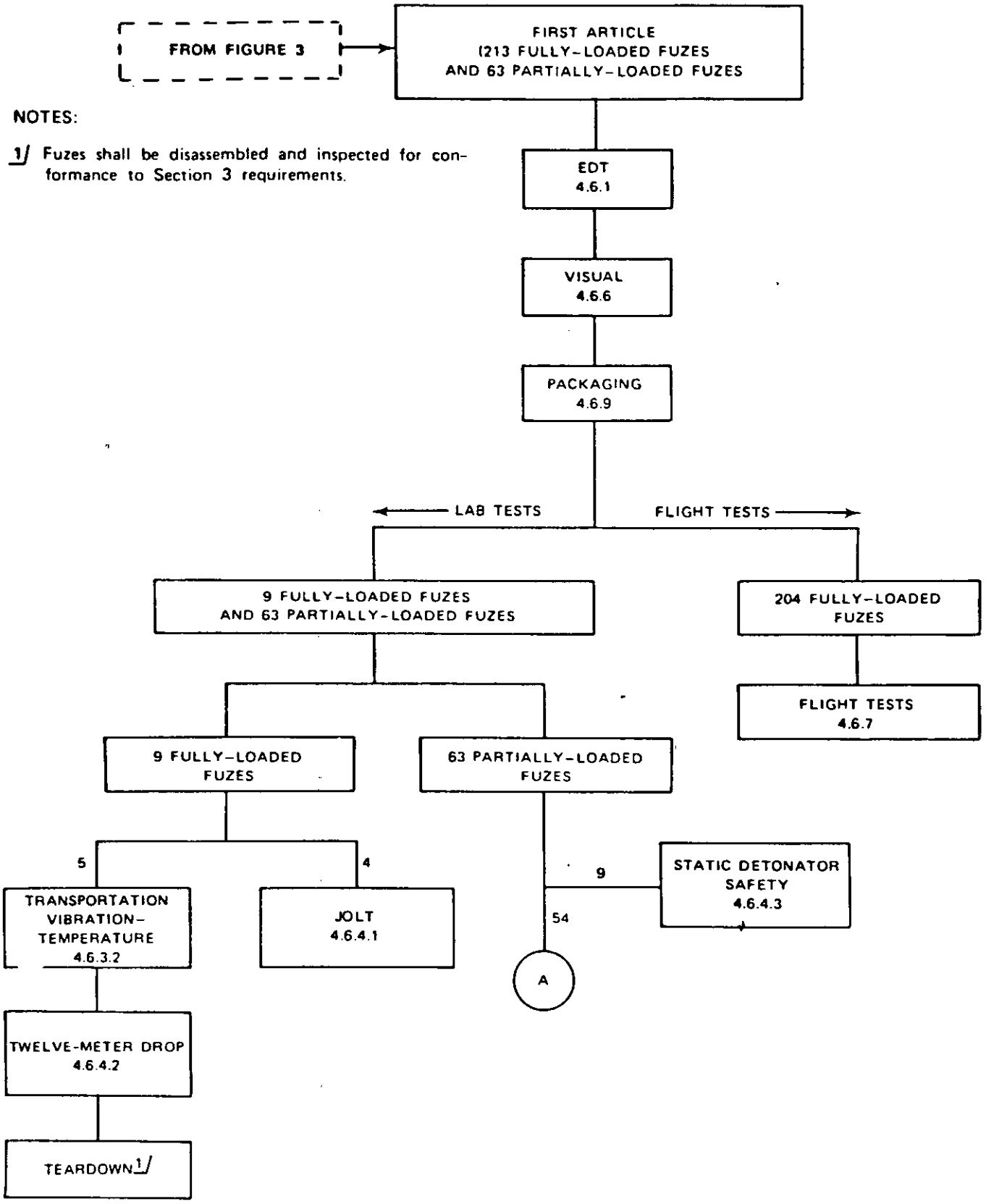


FIGURE 4. First article inspection sequence.

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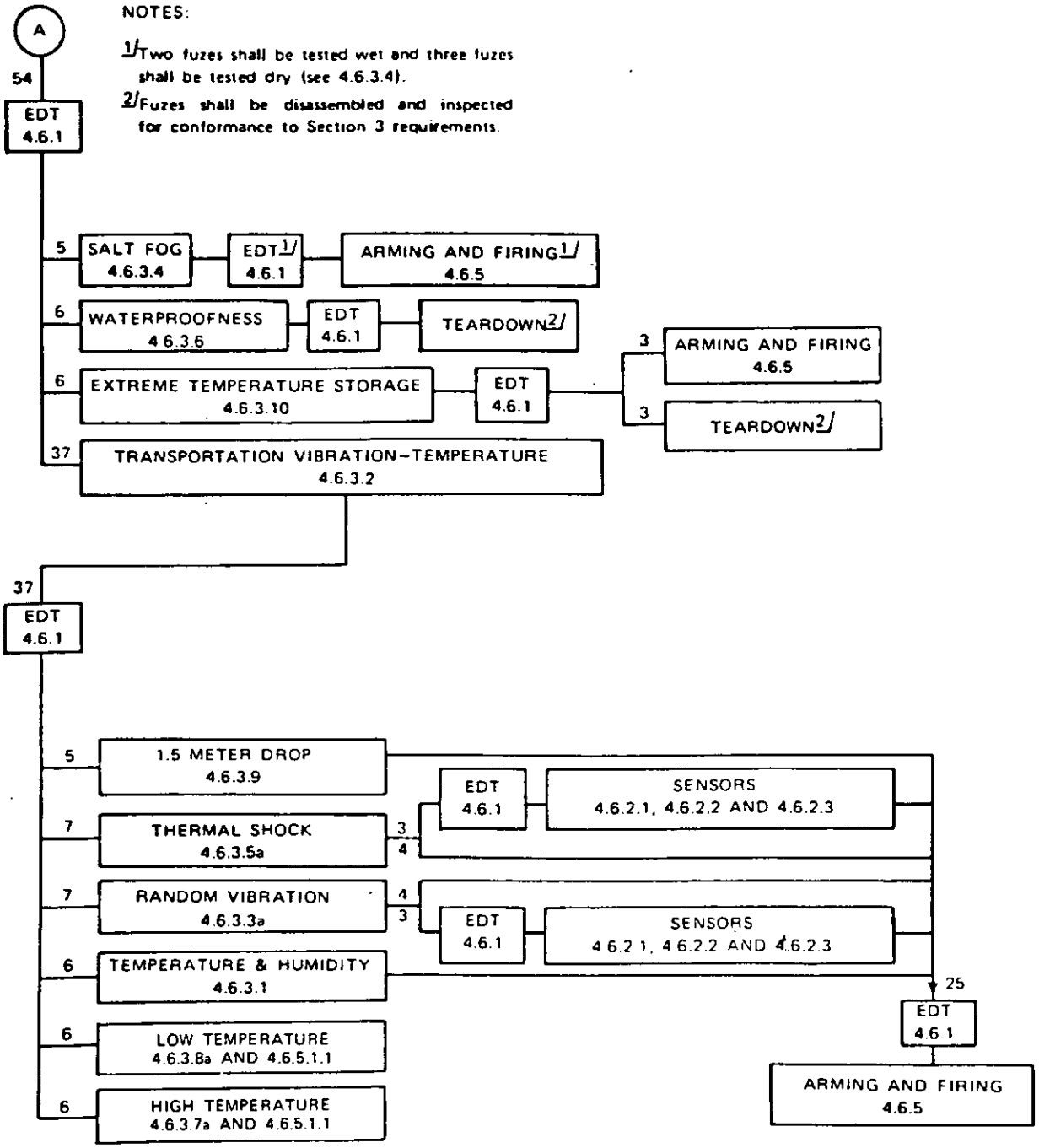
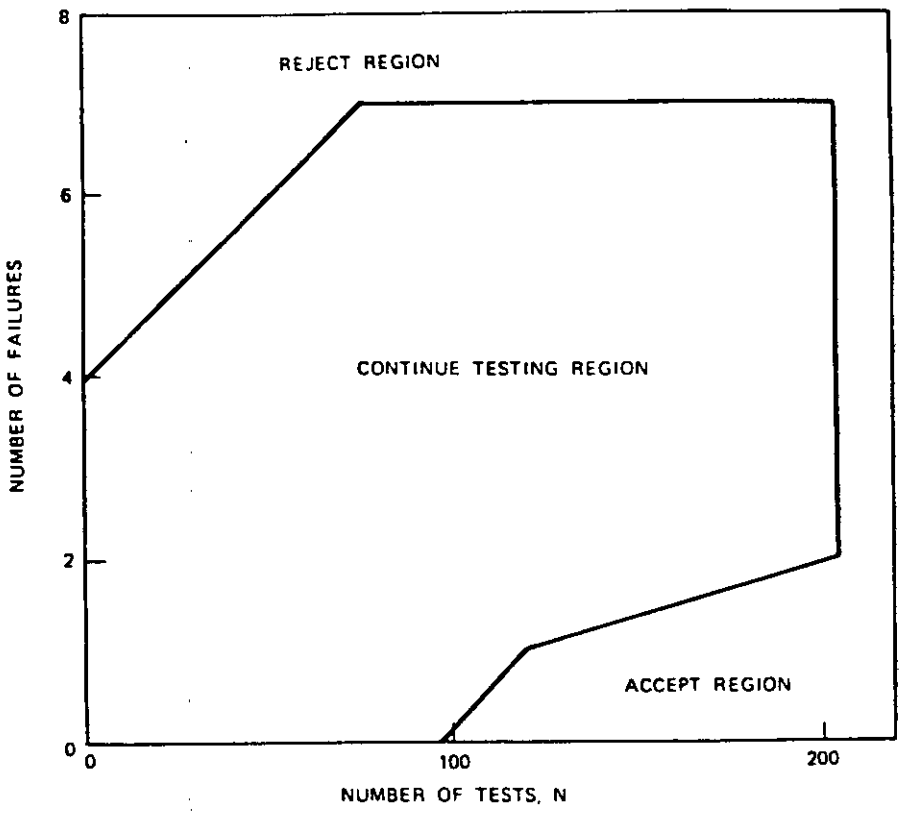


FIGURE 4. First article inspection sequence. - Continued.

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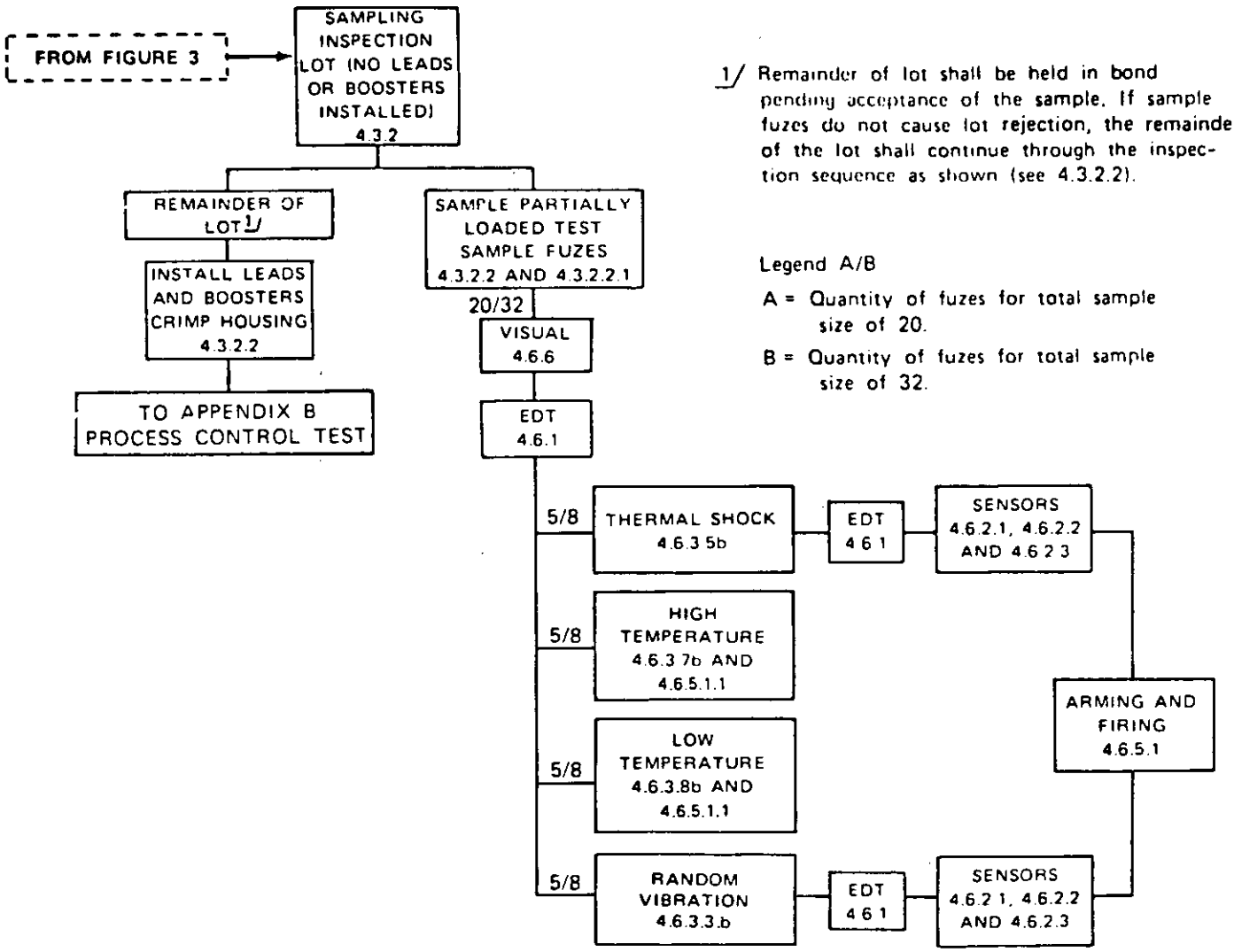


NUMBER OF FAILURES	ACCEPT IF N ≥	REJECT IF N ≤
0	97	0
1	121	0
2	204	0
3	204	0
4	204	0
5	204	25
6	204	51
7	204	76

NOTE. Reject on 8 failures regardless of value of N.

FIGURE 5. Flight test acceptance and rejection criteria.

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1/ Remainder of lot shall be held in bond pending acceptance of the sample. If sample fuzes do not cause lot rejection, the remainder of the lot shall continue through the inspection sequence as shown (see 4.3.2.2).

Legend A/B

A = Quantity of fuzes for total sample size of 20.

B = Quantity of fuzes for total sample size of 32.

FIGURE 6. Sampling inspection lot inspection sequence.

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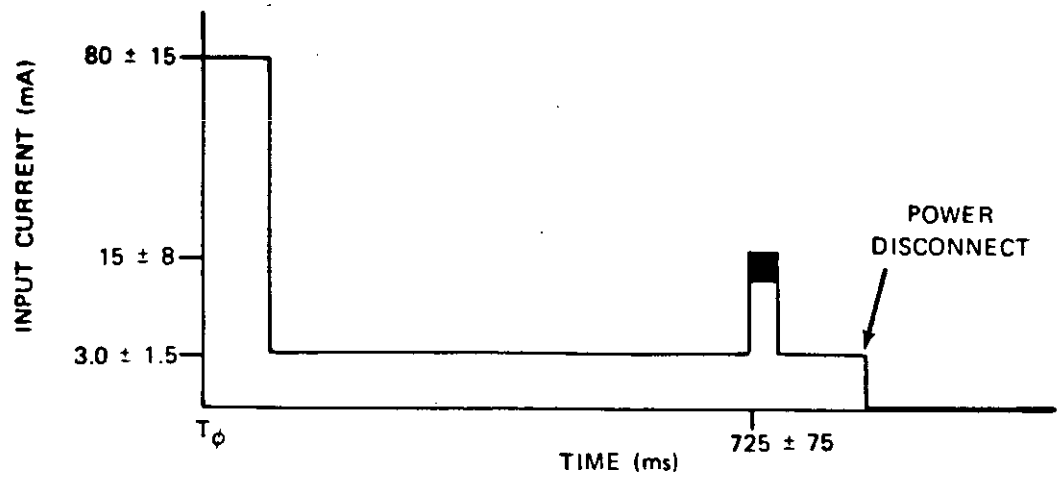


FIGURE 7. Navy mode test current waveform.

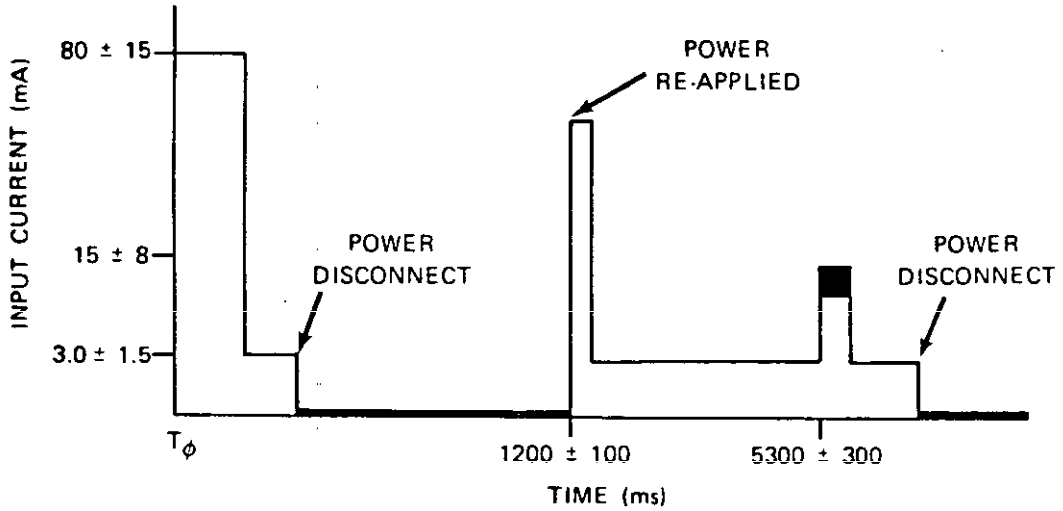


FIGURE 8. Navy mode interruped power test current waveform.

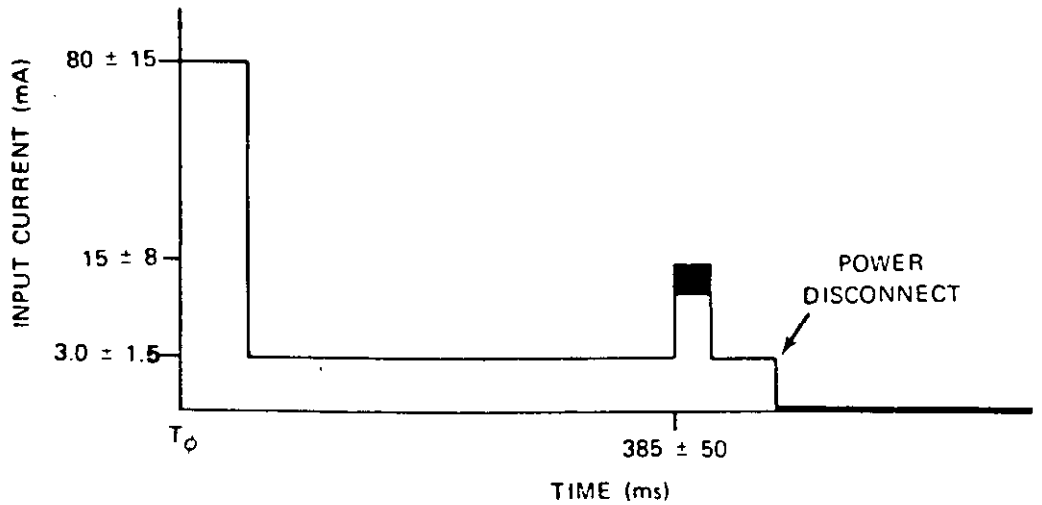


FIGURE 9. Air Force mode test current waveform.

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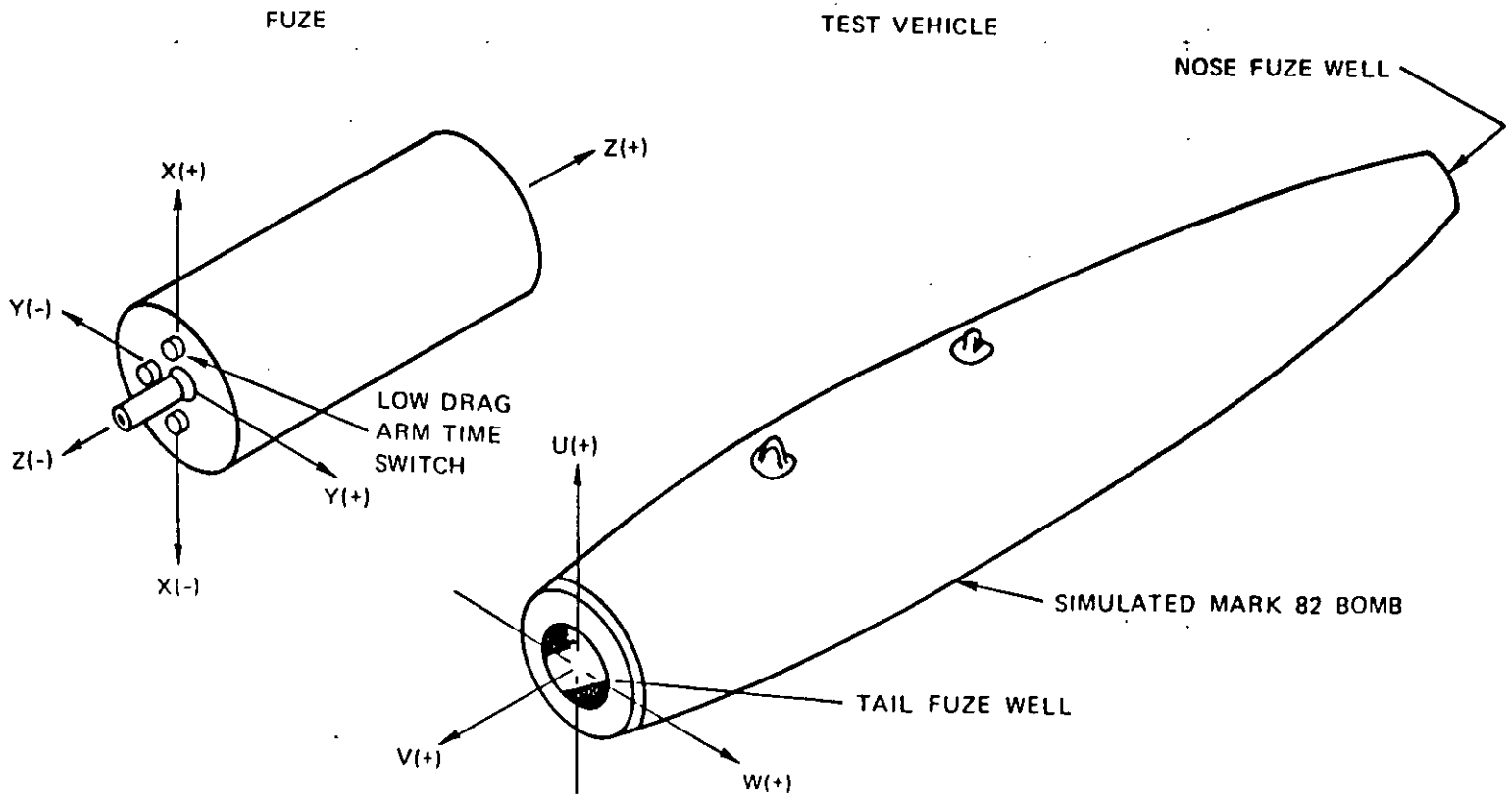


FIGURE 10. Definition of fuze axes and test vehicle axes.

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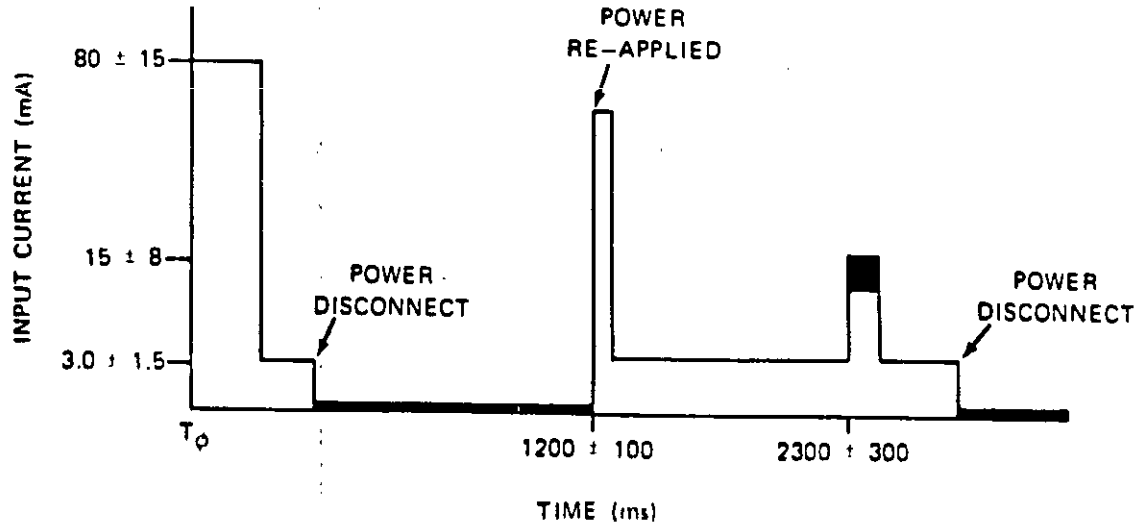


FIGURE 11. Retard sensors, high drag test current waveform.

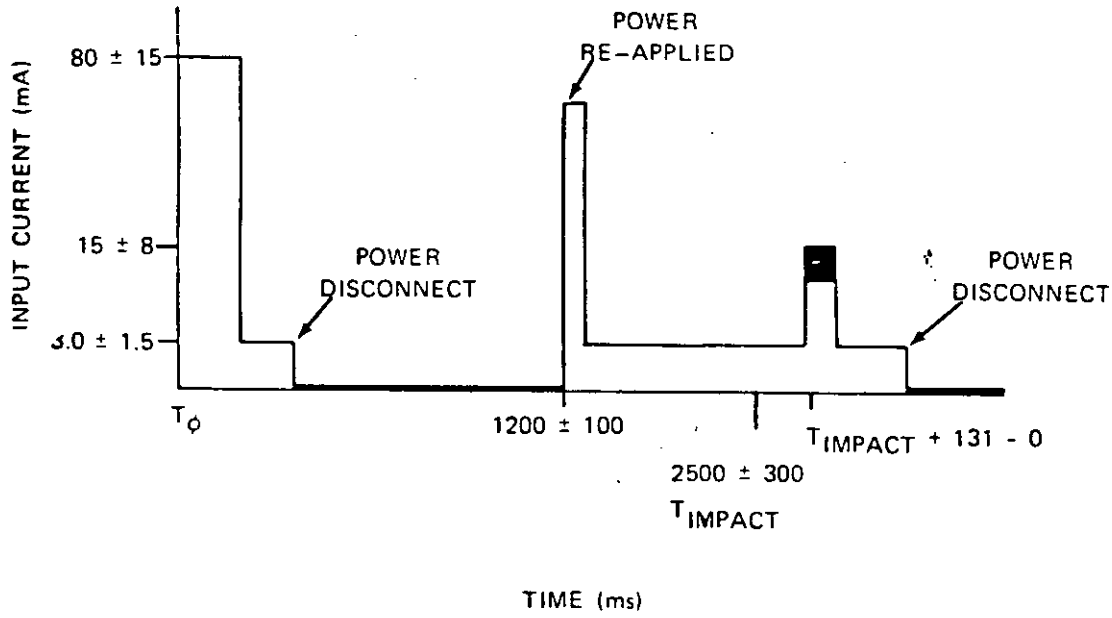


FIGURE 12. Impact sensors test current waveform.

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APPENDIX A

DRAWING CONFORMANCE INSPECTIONS

10. SCOPE

10.1 Scope. This appendix details the drawing conformance inspections required on each lot of parts, subassemblies, and assemblies of the fuze during manufacture, and on the completely assembled fuze. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. INSPECTIONS

30.1 Detailed inspection. The following drawing conformance inspections shall be performed in accordance with MIL-STD-105, Inspection Level II, AQL of 1.0 percent defective for all MAJOR defects. All CRITICAL characteristics shall be 100-percent inspected.

30.1.1 Drawing 1379AS100.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Part missing	Visual
CRITICAL 2	Feature not visible	Visual

30.1.2 Drawing 1379AS117.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Surface finish out-of tolerance	Gauge
CRITICAL 2	Hardness improper	Certification

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30.1.3 Drawing 1379AS157.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Malfunction	Electrical
MAJOR 101	Malfunction	Electrical

30.1.4 Drawing 1379AS164.

Classification of defect	Characteristic	Method of inspection
MAJOR 101	Material incorrect	Certification
MAJOR 102	Hardness incorrect	Certification
MAJOR 103	Plating out-of-tolerance	Gauge
MAJOR 104	Dimension out-of-tolerance	Gauge

30.1.5 Drawing 1379AS703.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Feature mislocated	Visual

30.1.6 Drawing 1379AS706.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Feature mislocated	Gauge
CRITICAL 2	Feature mislocated	Gauge
MAJOR 101	Feature mislocated	Gauge
MAJOR 102	Feature mislocated	Gauge
MAJOR 103	Feature mislocated	Gauge
MAJOR 104	Dimension out-of-tolerance	Gauge
MAJOR 105	Dimension out-of-tolerance	Gauge

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30.1.7 Drawing 1379AS707.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Dimension out-of-tolerance	Gauge
CRITICAL 2	Dimension out-of-tolerance	Gauge
MAJOR 101	Feature mislocated	Gauge
MAJOR 102	Feature mislocated	Gauge
MAJOR 103	Feature mislocated	Gauge
MAJOR 104	Feature mislocated	Gauge
MAJOR 105	Feature mislocated	Gauge
MAJOR 106	Dimension out-of-tolerance	Gauge
MAJOR 107	Feature mislocated	Gauge
MAJOR 108	Feature mislocated	Gauge
MAJOR 109	Feature mislocated	Gauge

30.1.8 Drawing 1379AS708.

Classification of defect	Characteristic	Method of inspection
MAJOR 101	Dimension out-of-tolerance	Gauge
MAJOR 102	Feature mislocated	Gauge
MAJOR 103	Dimension out-of-tolerance	Gauge
MAJOR 104	Feature mislocated	Gauge
MAJOR 105	Dimension out-of-tolerance	Gauge

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30.1.9 Drawing 1379AS710.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Feature not visible	Visual
CRITICAL 2	Part mislocated	Visual
MAJOR 101	Part mislocated	Visual
MAJOR 102	Value out-of-tolerance	Electrical
MAJOR 103	Feature mislocated or unbonded	Visual
MAJOR 104	Feature mislocated	Visual/ mechanical

30.1.10 Drawing 1379AS720.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Resistance to movement	Visual
CRITICAL 2	Part missing	Visual
CRITICAL 3	Part missing	Visual
CRITICAL 4	Feature mislocated	Visual

30.1.11 Drawing 1379AS730.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Operation not performed	Record
CRITICAL 2	Malfunction	Electrical
MAJOR 101	Dimension out-of-tolerance	Gauge
MAJOR 102	Dimension out-of-tolerance	Mechanical
MAJOR 103	Dimension out-of-tolerance	Mechanical

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30.1.12 Drawing 1379AS732.

Classification of defect	Characteristic	Method of inspection
CRITICAL 1	Dimension out-of-tolerance	Gauge
MAJOR 101	Dimension out-of-tolerance	Gauge
MAJOR 102	Feature mislocated	Gauge
MAJOR 103	Dimension out-of-tolerance	Gauge

30.1.13 Drawing 1379AS735.

Classification of defect		Method of inspection
MAJOR 101	Torque out-of-tolerance	Mechanical

30.1.14 Drawing 1379AS741.

Classification of defect	Characteristic	Method of inspection
MAJOR 101	Feature missing	Gauge

30.1.15 Drawing 1379AS746.

Classification of defect	Characteristic	Method of inspection
MAJOR 101	Dimension out-of-tolerance	Gauge

30.1.16 Drawing 1379AS747.

Classification of defect		Method of inspection
MAJOR 101	Dimension out-of-tolerance	Mechanical
MAJOR 102	Dimension out-of-tolerance	Mechanical

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30.1.17 Drawing 1379AS748.

Classification of defect	Characteristic	Method of inspection
MAJOR 101	Dimension out-of-tolerance	Certification
MAJOR 102	Dimension out-of-tolerance	Gauge
MAJOR 103	Part out-of-tolerance	Mechanical

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APPENDIX B

PROCESS CONTROL TEST

10. SCOPE.

10.1 Scope. The purpose of this appendix is to describe the Process Control Test (PCT) that is identified on Figure 6 of this specification. Since the fuze contains explosive components that make normal inspection testing hazardous, the acceptance testing has been separated into two parts: the sampling inspection that is performed on partially loaded fuzes, and the PCT that is performed on fully loaded fuzes. The sampling inspection is described in the main body of this specification. The PCT is described in this appendix. The purpose of the PCT is to verify the final assembly process in which the explosive lead and booster are installed and the fuze housing crimped as identified on Drawing 1379AS100. This is accomplished by performing inspections during the manufacturing process as described below. The PCT shall consist of an Electronic Diagnostic Test, Leak Test, Weight Verification, Visual Inspection, and a Fuze Firing Test that is conducted on a sampling basis.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DESCRIPTION OF TESTS.

30.1 Electronic diagnostic test (100 percent). An Electronic Diagnostic Test (EDT) shall be performed on each fuze to verify conformance with Drawing 1379AS100. Fuzes that fail EDT shall be rejected on an individual basis.

30.2 Leak test (100 percent). A leak test shall be performed on each fuze to verify conformance to paragraph 3.2.3.11. This test shall be performed with the shipping plug removed. Fuzes that fail leak test shall be rejected on an individual basis.

30.3 Weight verification (100 percent). Each fuze shall be weighed to verify that the explosive booster has been installed. Fuzes that do not contain a booster shall be rejected.

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30.4 Visual inspection (100 percent). A visual inspection shall be performed on each fuze to verify conformance to Drawing 1379AS100. Fuzes which do not meet the requirements of paragraph 3.2.6 shall be rejected on an individual basis.

30.5 Firing test (sample). On a sampling basis described below, fully loaded fuzes shall be fired. The equipment used to arm and fire the fuze shall be certified by the Government.

30.5.1 Firing. The firing test shall be conducted in accordance with 4.6.5.2.

30.6 Selection of test samples.

a. Prior to production or after a break in production exceeding 90 days, the first three fuzes processed shall be fired to verify the initial process.

b. One sample fuze from each subsequent batch shall be randomly selected after the batch has been completely processed.

30.7 Batches.

a. Fuzes shall be processed in batches, assembled from components processed in batches.

b. The batch size starts at 80 fuzes and increases in the following steps as follows: 80, 150, 225, 335, 500 and 675.

c. The initial batch size shall be 80 fuzes. The batch size may be increased in steps when the samples from the preceding 10 batches have been tested and there have been no PCT failures.

d. The maximum batch size shall be 675 fuzes.

30.8 Failure determination.

a. A failure of the fuze that is not associated with the loading process shall be considered a fuze failure.

b. A failure of the fuze that is associated with the loading process shall be considered to be a PCT failure.

c. When a PCT failure has occurred, a lower batch size shall be implemented immediately if the batch size exceeds 80 fuzes.

d. Failure analysis and corrective action shall be performed on each failed fuze when possible.

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e. If the fuze has armed and a proximity trigger source fails to fire the fuze, a hazardous situation exists. Impacting the fuze may trigger the impact sensor and detonate the fuze. Static electricity may initiate the detonator. Extreme caution shall be exercised. Failure analysis may not be possible.

30.9 Fuze failure.

a. When a fuze failure occurs, failure analysis and corrective action shall be implemented.

b. The batch size does not have to be reduced.

30.10 PCT failure. When a PCT failure has occured, the following action shall take place.

a. The next batch size shall be reduced, if the current batch size is greater than 80 fuzes.

b. Screen for the cause of failure.

c. Determine corrective action.

d. Implement corrective action.

e. If the fuze housing is opened the batch shall be resubjected to the PCT.

30.11 Batch acceptance and rejection criteria.

30.11.1 Batch acceptance criteria. If the sample fuze arms within the specified arm time and detonates the booster the batch shall be accepted.

30.11.2 Batch rejection criteria. If the sample fuze does not arm within the specified arm time or does not detonate the booster, and the failure is determined to be associated with the loading process, the batch shall be rejected and further sampling shall be required as follows:

a. Randomly select three additional sample fuzes from the same batch and subject them to a firing test.

b. If all three sample fuzes arm within the specified arm time, and detonate the booster, the batch shall be accepted.

c. If any one of the three sample fuzes does not arm within the specified arm time or does not detonate the booster, the batch shall be rejected.

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d. The failed batch and all subsequent batches are rejected until analysis has been performed and corrective action has been implemented.

e. Any safety defect shall cause immediate rejection of the batch and all other batches containing fuzes from the same sampling inspection lot.

30.12 Reporting.

30.12.1 Routine reporting.

a. A summary of data, failure analysis and corrective action shall be prepared for the government on all PCT tests performed.

b. Data shall include dates, quantity of fuzes processed, batch size, inspection lot number, tests performed, arming times, detonation results, fuze identification number of failed fuzes, failure analysis, and corrective action taken.

c. Data shall be submitted to the government in accordance with contract requirements (see 4.5f).

30.12.2 Urgent reporting.

30.12.2.1 The government shall be notified within 24 hours if test results indicate that there is a problem requiring immediate correction or any one of the following conditions exit:

a. Any two consecutive fuzes fail the firing test.

b. The last 30 sample fuzes have a total of 3 failures.

c. Three sequential failures in the same failure mode have occurred.

d. Any critical failure or safety problem.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-85571A(AS)	FUZE, BOMB, ELECTRONIC, FMU-139/B

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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